

RESQ-LINK

Tuguegarao City Command Center System

1. Introduction

The Tuguegarao City Command Center System is a web-based emergency response platform designed to modernize how the city handles incidents reported by the public. It provides a centralized system for recording, analyzing, and dispatching resources to emergencies using modern web technologies, AI-enhanced tools, and integrated communication channels. The system is designed with scalability, mobile accessibility, and real-time data transmission in mind, enabling faster, more informed responses by the city's first responders.

2. Key Actors and Roles

- Callers: Individuals reporting an emergency, either through the web application, SMS, or voice call.
- Dispatchers: Personnel at the command center who manage, classify, and assign incoming incidents.
- Responders: Field personnel such as paramedics, firefighters, or police who attend to the emergency.
- Supervisors/Admins: Users responsible for reviewing logs, managing access, and overseeing system health.

3. End-to-End Operational Flow

1. Caller submits an emergency report using the web app, SMS, or phone call.
2. The Command Center System receives the input and determines the input method:
 - SMS is parsed and tagged for urgency.
 - Calls are transcribed using an AI engine (e.g., Whisper API, OpenAI, Gemini).
 - Forms are automatically formatted and added to the emergency queue.
3. An AI suggestion engine categorizes the report (e.g., medical, fire, police) and provides possible responders.
4. Dispatchers verify, assign the nearest responders based on live GPS, and push incident info via mobile notification, SMS Notification, Voice Call.
5. Responders receive alerts on their Capacitor-based mobile app, which includes:
 - Caller's reported issue
 - Live map with route to incident

- Status update buttons (e.g., En Route, On Scene)
6. The system logs all activities and updates in real time. When resolved, the incident is closed and archived.

4. Technical Architecture and Stack

4.1 Frontend (Web and Mobile)

- Vue.js or React used to build a responsive Web App.
- Capacitor wraps the Web App into installable apps for Android and iOS.
- Features include offline caching, push notifications, camera access, geolocation, and call intents.

4.2 Backend (Laravel API)

- Laravel framework provides RESTful APIs for web and mobile clients.
- Authentication using Laravel Sanctum or Passport.
- Manages emergency queue, caller database, responder records, and logs.
- Handles role-based permissions, activity logs, and SMS notifications.
- Admin dashboard built using Laravel Blade or Inertia.js.

4.3 Communication Channels

- Voice Call Transcription: Handled by Whisper API or Twilio Programmable Voice.
- SMS Integration: Handled via GSM modem (AT commands) or third-party API (Globe, Twilio).
- Push Notifications: Firebase Cloud Messaging used for live updates to responders.

4.4 Real-Time Updates

- Implemented using Laravel Echo + Pusher or a Socket.IO Node.js microservice.
- Used for real-time location updates, new incident alerts, and chat between dispatch and responders.

4.5 Location & Map Services

- Capacitor Geolocation Plugin for live responder tracking.
- Google Maps API (or Leaflet + OpenStreetMap) used to display routes and incident points.
- Backend selects nearest responders based on a distance algorithm.

5. Data Flow Lifecycle

1. Caller initiates report.
2. Input flows into Laravel API.
3. System validates, stores, and timestamps the report.
4. AI module evaluates urgency and context.
5. Dispatcher verifies and assigns responder from real-time responder pool.
6. Updates and status changes flow back to the database, notifications sent.

- 7. Responder actions recorded, arrival and completion timestamps logged.
- 8. System archives incident, makes it available in reporting dashboard.

6. Security and Access Control

- HTTPS enforced across all communications.
- JWT or token-based authentication for mobile clients.
- Roles and permissions managed via middleware.
- Rate limiting and input validation to prevent abuse/spam.
- Logs are encrypted and protected against tampering.

7. Reporting and Analytics

- Daily/weekly/monthly report generation for incidents.
- Metrics: average response time, most frequent incident type, busiest barangay, etc.
- Exportable in Excel or PDF formats.
- Graphical dashboards available to admins and supervisors.

8. Deployment and Scalability

- Laravel and MySQL hosted on cloud (e.g., DigitalOcean, AWS EC2).
- NGINX used as reverse proxy.
- Load balancing and read-replicas available for horizontal scaling.
- Capacitor apps deployed to Google Play and Apple App Store.
- CI/CD via GitHub Actions or GitLab Pipelines.